

ABOUT

↑Sagar Panchal
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Engineer with system-level expertise in automation, robotics, machine software and embedded systems. I design reliable, production-grade industrial solutions integrating motion control, sensors, firmware and real-time logic. Focused on roles requiring ownership, architectural thinking and end-to-end delivery of intelligent robotic systems.

EDUCATION

MS, COMPUTER ENGINEERING (3.41/4)	– DEC-17	🎓
NEW YORK UNIVERSITY		🌳 NYU
BE, INSTRUMENTATION & CONTROLS ENGINEERING (7.04/10)	– JUN-14	🎓
GUJARAT TECHNOLOGICAL UNIVERSITY		

SKILLS

PROGRAMMING:	C/C++, CS, JAVA(PRIMARY), PYTHON (AUTOMATION AND AI SCRIPTING), WEB(HTML, CSS, JS), DATABASES(SQL, MDB)
AUTOMATION & ROBOTICS:	REAL-TIME SYSTEMS (VxWORKS), MV3100 INDUSTRIAL CONTROLLER, MOTION CONTROL (ROTARY, CARTESIAN, SMT AUTOMATION SYSTEMS), MACHINE VISION (COGNEX, KEYENCE, OPENCV), MES INTEGRATION (MANUFACTURING WORKFLOWS)
SCM & TOOLS:	GIT (BASH & SOURCETREE), JIRA WORKFLOWS, VS CODE, VISUAL STUDIO (2010, 15, 22), NETBEANS, WIRESHARK, WINDRIVER, TORNADO

PROFESSIONAL EXPERIENCE

🧰 UNIVERSAL INSTRUMENTS, NY

👤 SOFTWARE ENGINEER III - AUTOMATION

Sep 22 - present

Core Responsibilities:

- Maintain and enhance high-precision, high-speed software modules for Fuzion SMT Platform.
- Provide remote production-line support by investigating issues, analyzing logs, replicating site conditions to perform root-cause analysis, reviewing configurations and delivering software fixes, patches, and release updates.
- Enable field and application engineers to correctly set up and validate applications by offering technical guidance, interpreting machine data and supplying required configuration or software updates.

Key contributions:

- Built the SBS feature integrating Keyence SR-X300, delivering Fuzion's first side-barcode scanning capability with live-view, traceability, and error-recovery support.
- Upgraded the AMS workflow from one point to 13 locations on wafer, delivering a major accuracy improvement in wafer placement and calibration.
- Built the wafer-to-board mapping system that converts customer wafer maps into Fuzion board templates, enabling direct use of wafer data and eliminating hours of manual template creation.
- Delivered Selective Placement, enabling Fuzion's first smart pick/buffer/place workflow using DFF-based categories and avoiding-priority rules, improving overall product quality.
- Implemented AVP, adding user-defined fiducial capability beyond fixed shapes and improving alignment flexibility across ECS, NPI, and Production workflows.
- Integrated Cognex QuickBuild (VPP) with MCOS↔QBIT TCP/IP triggers and full reject/retry handling, enabling Dispense Quality Inspection on the Fuzion Precision-Bonder platform for the first time.
- Delivered key platform enhancements — Light Intensity Controller, nested-tray support, isosceles fiducial, and configurable park-position — among many other features across Fuzion.

🧰 NOR-CAL CONTROLS, CA

☐ AUTOMATION ENGINEER

Jan 22 - Jul 22

- Designed SCADA interfaces and integrated PLC logic with inverter and sensor data to deliver reliable solar power-plant automation.

🧰 MINI-CIRCUITS, NY

☐ SOFTWARE ENGINEER - MACHINE AUTOMATION

Feb 18 - Nov 21

- Developed and enhanced Automation Software for multi-function rotary systems, coordinating servo-motion via PCI-e interfaces and integrating sensor-driven workflows, computer-vision inspection and EVT test automation.
- Built a standalone laser-measurement station using precision sensing to classify mechanical-switch dimensions, improving QA accuracy and reducing manual effort.

🧰 PRIMA AUTOMATION, AHMEDABAD, INDIA

☐ AUTOMATION ENGINEER

Jun 15 - Jan 16

- Programmed Siemens PLCs and developed HMI interfaces for machine-level automation involving motion control and sensor integration and industrial communication protocols.

ACADEMIC WORK EXPERIENCE

📖 REVMAX FLEET OPTIMIZATION, NY BASED STARTUP

- DEVELOPER INTERN

Fall 2017

- Built a BLE Doppler-radar pedestrian-counting device, covering firmware, sensor integration & field-deployment.

📖 NEW YORK UNIVERSITY, NY

- INSTRUCTOR (ITEST PROGRAM), ROBOTICS & MECHATRONICS DPT.

Fall 2017

- Mentored high-school robotics teams in Arduino-based programming, sensors and autonomous robot workflows.

- LEAD INSTRUCTOR (CREST PROGRAM), CENTER FOR K12 STEM EDUCATION

Summer 2017

- Led K-12 STEM instruction in robotics, Arduino, sensors and hands-on project building.